

# When Models Can't Follow: Testing Instruction Adherence Across 256 LLMs

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## Abstract

Despite the widespread deployment of Large Language Models in various applications, systematic evaluation of their instruction-following capabilities remains a critical challenge. While comprehensive benchmarks exist, there is value in focused, reproducible assessments that can quickly diagnose specific instruction adherence patterns. Additionally, as newer models may be trained on existing benchmark datasets, novel evaluation approaches are needed to assess genuine instruction-following capabilities rather than memorized benchmark performance. This paper presents a streamlined evaluation framework using twenty carefully designed prompts to assess LLM instruction-following performance across diverse task categories. We demonstrate this framework through a large-scale empirical study conducted on October 14, 2025, testing 256 verified working models from the 331 models available via OpenRouter. To ensure methodological rigor and prevent selection bias, we first verified each model's basic functionality with a simple factual query before inclusion in the instruction-following evaluation. Unlike large-scale benchmarks that may require extensive computational resources, our approach offers a practical diagnostic tool that researchers and practitioners can readily apply to evaluate new models or compare existing ones.

Our methodology builds upon the concept of verifiable instructions while introducing a compact test suite that balances comprehensiveness with efficiency. Each prompt in our framework targets distinct aspects of instruction following, including format compliance, content constraints, logical sequencing, and multi-step task execution. We evaluate 256 verified working models from major providers (OpenAI, Anthropic, Google, Meta, Mistral) and emerging implementations (Qwen, DeepSeek, community models) using this framework and provide a comparative analysis of their performance patterns. Our findings reveal consistent failure modes across models and identify specific instruction types that pose particular challenges. This work contributes both a practical evaluation tool and one of the most comprehensive empirical analyses of instruction-following capabilities across the contemporary LLM landscape, offering insights into the current limitations of instruction following in language models.

## Introduction

Large Language Models have emerged as transformative technologies across numerous domains, demonstrating remarkable capabilities in natural language understanding, generation, and reasoning tasks. Among these capabilities, the ability to accurately follow natural language instructions stands as a fundamental requirement for reliable deployment in real-world applications. This instruction-following capability determines whether models can execute user requests precisely, maintain consistency with specified constraints, and produce outputs that align with intended objectives. However, despite the critical importance of instruction adherence, systematic evaluation of this capability

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remains challenging due to the inherent ambiguity and subjectivity of natural language interpretation. As Zhou et al. [25] demonstrate in their comprehensive evaluation framework, the assessment of instruction-following abilities requires careful consideration of verifiable metrics and reproducible methodologies to overcome the limitations of subjective human evaluation and potentially biased model-based assessments. The need for practical, focused evaluation approaches that can efficiently diagnose instruction-following performance across diverse model architectures motivates the development of streamlined diagnostic tools that complement existing comprehensive benchmarks.

The development of effective evaluation methodologies for instruction-following capabilities has evolved along several distinct trajectories. Comprehensive benchmarks such as IFEval [25] have established rigorous frameworks utilizing hundreds of verifiable instructions across diverse categories, providing broad coverage of potential failure modes and instruction types. However, these extensive evaluation suites present practical challenges for researchers and practitioners who require rapid diagnostic capabilities or wish to conduct frequent comparative assessments across model iterations. The computational resources and time requirements for executing large-scale benchmarks can impede iterative development processes and limit accessibility for resource-constrained research environments. Furthermore, while comprehensive benchmarks excel at providing statistical robustness and broad coverage, they may obscure specific patterns of instruction-following failures that become apparent only through focused, targeted evaluation. This gap between comprehensive assessment and practical diagnostics suggests the need for complementary evaluation approaches that balance thoroughness with efficiency, enabling rapid identification of instruction-following weaknesses while maintaining sufficient rigor for meaningful model comparison.

## Related Work

### Comprehensive Evaluation Benchmarks

The landscape of language model evaluation has witnessed significant evolution, with several major benchmarks establishing themselves as standard assessment tools. MMLU (Massive Multitask Language Understanding) introduced by Hendrycks et al. [3] provides broad coverage across 57 subjects, testing models’ knowledge and reasoning capabilities. While comprehensive in scope, MMLU primarily focuses on knowledge retrieval rather than instruction adherence, making it complementary but distinct from instruction-following evaluation.

BIG-bench [15] represents another large-scale collaborative effort, comprising over 200 tasks designed to probe diverse model capabilities. The benchmark’s strength lies in its breadth and community-driven task development, though its heterogeneous nature can make it challenging to isolate specific capability deficits such as instruction-following failures. The sheer scale of BIG-bench, while valuable for comprehensive assessment, presents practical barriers for rapid iterative testing during model development.

### Instruction-Following Specific Evaluations

More targeted approaches to instruction-following evaluation have emerged to address the specific challenges of assessing instruction adherence. IFEval [25] stands as the most directly relevant prior work, introducing a comprehensive framework with verifiable instructions that enable objective assessment. Their methodology emphasizes the

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importance of eliminating subjective judgment from evaluation, a principle we adopt and extend in our more focused framework.

Recent benchmarks have further refined instruction-following evaluation. InFoBench [12] provides a multi-level evaluation framework that assesses both content and format compliance through its DRFR (Decomposed Requirements Following Ratio) metric, which decomposes complex instructions into simpler verifiable criteria. FollowBench [4] introduces fine-grained constraint categories to analyze specific failure modes across 820 samples. PandaLM [18] offers an automatic evaluation benchmark specifically designed for optimizing instruction tuning, demonstrating the growing recognition of instruction adherence as a critical optimization target.

Specialized benchmarks have emerged for specific instruction-following scenarios. SIFo [1] focuses specifically on sequential instruction following in open-domain settings, while EifBench [27] tackles extremely complex instruction-following scenarios. CELLO [2] extends evaluation to Chinese language with 523 samples, addressing the need for multilingual instruction-following assessment. LLMBAR [24] provides meta-evaluation capabilities with both natural and adversarial test sets. RewardBench [6] evaluates reward models, which are crucial for instruction-following alignment through reinforcement learning.

The Self-Instruct framework [17] explores instruction-following from a different angle, focusing on models' ability to generate and follow their own instructions. This approach has inspired several instruction data generation methods including WizardLM's Evol-Instruct [22], which evolves instructions to increase complexity, Magpie [23], which synthesizes alignment data from scratch, and instruction backtranslation [8], which generates instructions from responses. While these self-referential approaches provide insights into instruction understanding, they differ from our focus on adherence to externally provided, precisely specified constraints.

Prometheus 2 [5] represents another evaluation-focused model, specifically trained to assess other language models' outputs. This meta-evaluation approach, along with Auto-J [7], highlights the complexity of instruction-following assessment and the potential for specialized evaluation models to complement benchmark-based approaches. Recent work has also extended instruction-following evaluation to retrieval tasks, with InfoSearch [26], FollowIR [20], and InstructIR [10] demonstrating that instruction adherence challenges extend beyond text generation.

## Prompt Engineering and Robustness Studies

Research into prompt engineering has revealed significant variability in model responses based on instruction phrasing and structure. Wei et al. [19] demonstrated that chain-of-thought prompting can improve complex reasoning, though it may introduce verbosity that conflicts with concise output requirements. This tension between reasoning transparency and instruction compliance motivates several of our test designs.

Robustness studies by Ribeiro et al. [13] and others have shown that models often fail on slight variations of tasks they seemingly master, suggesting that true instruction-following capability requires more than pattern matching. Our framework incorporates this insight by designing prompts that test genuine understanding rather than memorized responses.

## Complex Instruction Scenarios

Recent work has explored increasingly sophisticated instruction-following scenarios. Wen et al. [21] introduce benchmarks for complex instruction-following with multiple constraint composition, revealing that models often fail when multiple constraints must